Multitask Learning and Reinforcement Learning for Personalized Dialog Generation: An Empirical Study

Min Yang, Weiyi Huang, Wenting Tu, Qiang Qu, Ying Shen, and Kai Lei

***Abstract*— Open-domain dialog generation, which is a crucial component of artificial intelligence, is an essential and challenging problem. In this article, we present a personalized dialog system, which leverages the advantages of multitask learning and reinforcement learning for personalized dialogue generation (MRPDG). Specifically, MRPDG consists of two subtasks: 1) an author profiling module that recognizes user characteristics from the input sentence (auxiliary task) and 2) a personalized dialog generation system that generates informative, grammatical, and coherent responses with reinforcement learning algorithms (primary task). Three kinds of rewards are proposed to generate high-quality conversations. We investigate the effectiveness of three widely used reinforcement learning methods [i.e., Q-learning, policy gradient, and actor-critic (AC) algorithm] in a personalized dialog generation system and demonstrate that the AC algorithm achieves the best results on the underlying framework. Comprehensive experiments are conducted to evaluate the performance of the proposed model on two real-life data sets. Experimental results illustrate that MRPDG is able to produce high-quality personalized dialogs for users with different characteristics. Quantitatively, the proposed model can achieve better performance than the compared methods across different evaluation metrics, such as the human evaluation, BiLingual Evaluation Understudy (BLEU), and perplexity.**

***Index Terms*— Author profiling, multitask learning, personalized dialog generation, reinforcement learning.**

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I. Introduction

BUILDING a dialog system that is capable of providing fluent, informative, and coherent responses is a long-term goal of artificial intelligence (AI). The previous work [1] revealed that personalized conversational systems could speed up the information retrieval process and contribute to a more positive perception of the interaction. In general, there are two types of dialog generation systems: the task-oriented dialog system and the open-domain dialog system. It is noteworthy that, in this article, we concentrate on the open-domain dialog system, which is based on the foundation of massive corpora of real-world human dialogs with no hand-crafted rules.

Inspired by the remarkable progress of the encoder–decoder framework in neural machine translation and image captioning, many dialog systems adopt sequence-to-sequence (seq2seq) model to generate dialogs [2], [3]. Typically, the seq2seq model first employs a recurrent neural network (RNN) encoder to encode the conversation history as a vector, which is then fed into an RNN decoder to produce responses word by word. The encoder–decoder-based models have dominated the study of dialog generation since these models can be trained end-to-end and scale to very large training data, which brings the results of good flexibility and quality. Despite the effectiveness of previous studies, there are still several technical challenges for generating personalized, coherent, informative, and fluent conversations.

Previous studies [1], [4] revealed that adapting dialog strategy in response to user characteristics (e.g., age, gender, education, and location) improves the usability of dialog systems, especially for novice users. The user characteristics are of primary importance in building an effective persuasion dialog system since they are directly connected to the subject and the speaking style of users' responses. The user characteristics can impact the process of chatting. Demonstrably, infusing user characteristics into dialog systems could enhance user satisfaction and contribute to highly applicable open-domain dialog systems. Despite their usefulness, to date, individual characteristics have gained little attention in encoder–decoder-based models that obtain the state-of-the-art results in dialog generation.

On the other hand, the typical seq2seq model is trained by predicting the next conversation utterance using the maximum likelihood estimation (MLE) objective function. However, the seq2seq model has two significant disadvantages.

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First, the evaluation metrics of dialog systems are not used when training the seq2seq model, which will magnify the differences between losses from training and testing stages. For example, the seq2seq model is typically trained by employing the cross-entropy strategy, while it is evaluated by applying nondifferentiable and discrete evaluation metrics, such as perplexity [2] and BiLingual Evaluation Understudy (BLEU) [5] at test time. Second, the standard MLE training will lead to the "exposure bias" problem [6] because of the discrepancy between the training and testing phases. Specifically, at training time, the seq2seq model predicts the next word conditioned on the previous ground-truth words. Nevertheless, during generation, the trained model is used to generate the next word by feeding the previously generated word back as input. Consequently, the seq2seq model is biased to only perform well on the ground-truth history distribution during training, and the errors may accumulate along the generated sequence at test time. The first several words of the generated response may be natural and reasonable, while the quality of the response deteriorates quickly. The previous work [7] showed that the MLE-based seq2seq models are likely to produce trivial, noncommittal, or similar responses that often involve high-frequency words or phrases along with the line of "*I do not know*."

In this article, the aforementioned limitations are considered and alleviated to some extent. We present a personalized dialog system, which leverages the advantages of **M**ultitask learning and **R**einforcement learning for **P**ersonalized **D**ialogue **G**eneration (MRPDG). MRPDG employs the sequence-to-sequence framework as its backbone, which uses a hierarchical long short-term memory (LSTM) to map the conversation history into a vector, and then uses an LSTM decoder to decode the target response word by word from the vector. In addition, MRPDG optimizes two related tasks simultaneously: author profiling (the auxiliary task) and personalized dialog generation (the primary task). The author profiling module, the first subtask in MRPDG, is trained with an LSTM encoder, which is shared across tasks. It helps to enhance the semantic comprehension of sentences and capture implicit user characteristics. More importantly, the author profiling (classification) results are also utilized to produce user-specific sentence representation that captures user characteristics, such as the speaking style and the background information. In the second subtask, an LSTM decoder is employed to generate the target response, treating the learned user-specific utterance representation as to the initial state of the LSTM decoder. In addition, the user characteristics are incorporated into the LSTM decoder directly as the extra input, alleviating the speaker inconsistency issue when generating responses. Different from the standard MLE training, we devise three kinds of rewards to delineate high-quality dialogs and employ reinforcement learning algorithms to guide MRPDG to produce highly rewarded dialogs. Thus, the proposed MRPDG model can overcome the nondifferentiable and "exposure bias" issues, which maximizes the expected reward for a conversation.

The main contributions of this article can be summarized as follows.

1) We leverage the author profiling task to learn better user-specific sentence representation for personalized dialog generation. The learned user-specific sentence representation can capture the characteristics of users with the help of author profiling tasks. In addition, the user characteristics are formulated as distributed representations, and they are fed into the decoder directly as an accessional input.
2) We propose MRPDG, which combines the power of the attention-based seq2seq model in learning semantic meanings of the conversations and the power of reinforcement learning in maximizing the long-term rewards over the conversations. Three types of rewards are proposed to generate natural, personalized dialogs.
3) We investigate the performance of three widely used reinforcement learning methods [i.e., Q-learning, policy gradient, and actor-critic (AC) algorithm] for personalized dialog generation, and experimentally demonstrate that the AC algorithm (i.e., MRPDG-AC) achieves the best results for generating personalized dialogs.
4) Experimental results on two real-life dialog data sets show that MRPDG achieves substantial improvements over the compared methods. A series of experiments are conducted to demonstrate the effectiveness of the proposed method from various perspectives, such as perplexity, BLEU, and human evaluation.

The remainder of this article is structured as follows. In Section II, we discuss the related studies for dialog generation. Section III defines the problem and briefly introduces the architecture of MRPDG. Section IV elaborates on the author profiling model. In Section V, we describe the proposed MRPDG model in detail. In Section VI, we set up the experiments. We illustrate and analyze the experimental results in Section VII. Section VIII concludes this article and presents some possible future work.

II. Related Work

The goal of dialog generation [8] is to generate natural, informative, and coherent dialog responses conditioned on the conversation history. It is an important but arduous task, which has received increasing attention recently. The articles [9] and [10] introduced a large number of previous techniques and data sets for building dialog systems.

A. Open-Domain Dialog Generation

Inspired by the remarkable progress of the encoder–decoder framework in text generation, a large number of studies have been proposed to generate responses using the seq2seq model [2]–[4], [11]. These models are end-to-end trainable and have good language modeling ability. L. Shang *et al.* [3] is one of the early models, which employs the encoder–decoder framework [12] to build conversation systems. The main idea is to first employ an LSTM encoder to encode the conversation history, which produces the distributed utterance representation with the fixed dimension. Then, an LSTM decoder is employed to produce the target response word by word,

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which uses the encoded utterance representation as to the initial state of the decoder. Subsequently, various techniques have been proposed to improve the quality of generated responses from different perspectives, such as diversity promotion [13], unknown words handling [14], prototype editing [15], and retrieval-based ensemble [16]. However, a well-known problem of these models is that they are prone to produce universal and even meaningless responses.

Recently, combining the encoder–decoder framework and reinforcement learning is demonstrated to be effective in boosting the overall performance of dialog systems. For example, [17] introduced a text generation strategy by combining the seq2seq model and Q-learning algorithm. A Q-network is trained to decide which word should be chosen from the candidates to amend the generated response. J. Li *et al.* [7] employed the policy gradient method [18] to generate highly rewarded dialogs that are informative, coherent, and easy to answer. D. Bahdanau *et al.* [19] generated text sequences using AC methods, which directly improved the test time metrics by using an additional critic network. Different from previous work, our model further enables the seq2seq framework adaptive to users' personalized dialog patterns with multitask learning.

B. Personalized Dialog Generation

Personalization techniques have shown increasing importance in user-centric applications [20], [21]. Recently, there were some studies being introduced to consider the individual characteristics and deal with the information conformance of users [1], [4], [22], [23]. For instance, [4] proposed a speaker–addressee model (SAM) for generating personalized dialogs, which encoded the interactive relations of two users by constructing persona vectors for each dialog. W.-N. Zhang *et al.* [1] proposed a transfer learning method to produce personalized dialogs, which first pretrained the encoder–decoder model on a large-scale dialog data set to learn general conversation patterns, and then fine-tuned the pretrained model with a small-scale personal dialog data set for personalized response generation. K. Mo *et al.* [22] worked on the task-specific personalized dialog system. The system employed the POMDP algorithm to capture the long-term rewards of the dialogs and designed a transfer learning scheme to learn personalized knowledge. M. Yang *et al.* [24] introduced a domain adaptation strategy for personalized response generation by employing the dual learning algorithm that extracted the reward signals for the reinforcement learning algorithm. Our approach differs from the above-mentioned methods in several aspects. First, we design three types of rewards to characterize high-quality dialogs. Second, we investigate three kinds of reinforcement learning methods and compare their performance on two real-life data sets. Third, we explore multitask learning to incorporate user characteristics into the seq2seq framework.

C. Multitask Learning

Multitask learning solves multiple related tasks jointly, aiming at boosting the overall performance of each individual task by leveraging useful information contained in multiple related tasks [25]–[27]. For example, [26] combined the advantages of multiple-domain classification tasks and information retrieval tasks. The joint model could take advantages of the large-scale cross-task data and the generalization ability by leveraging the comprehensive information across related tasks. M.-T. Luong *et al.* [27] integrated the seq2seq model with multitask learning, in which the encoder and decoder are shared by the related tasks. The experiments demonstrated that the translation performance could be improved by training on image captioning and syntactic parsing data. P. Liu *et al.* [28] learned some classification tasks (e.g., subjective classification and sentiment classification) simultaneously, which shares the text representation learning module. S. Venugopalan *et al.* [29] leveraged complex linguistic information in generating grammatical video captions with an extraneural language model. Y. Luan *et al.* [30] proposed a multitask learning approach to build a personalized dialog system by training dialog systems in response to different speakers jointly. R. Pasunuru and M. Bansal [31] enhanced the performance of video captioning by leveraging the language entailment generation as an auxiliary task. In this article, we share the LSTM encoder parameters of a seq2seq model and an author profiling model to incorporate the user characteristics through multitask learning.

III. Our Methodology

A. Problem Definition

In this section, we introduce the primary notations utilized in this article and formulate the learned tasks briefly. Recall that our task is to generate natural and personalized conversations given the conversation history. We represent a dialog between two users as a sequence of utterances $D = \{U_1, U_2, \ldots, U_K\}$, where K represents the number of utterances within a conversation. The kth utterance $U_k = \{w_{k,1}, w_{k,2}, \ldots, w_{k,N_k}\}$ contains a sequence of N_k words. Each word $w_{k,n}$ is chosen from a word vocabulary V, and $w_{k,n}$ indicates the nth word in the kth utterance. We define $U_{<k} = \{U_1, \ldots, U_{k-1}\}$.

For each user, we have the corresponding user profile (i.e., age, education, gender, location, and identity), denoted as y_k^{age}, $y_k^{\text{edu.}}$ y_k^{gender}, $y_k^{\text{loc.}}$, and y_k^{id}, respectively. These user profiles can be acquired from the registration of a dialog system. The gender is considered to be binary, e.g., either male or female, while the age is divided into five groups (10s: 10–19, 20s: 20–29, 30s: 30–39, 40s: 40–49, 50s: above 50). We consider the location to be the state or province as the user located. The education profile indicates the grade levels or educational stages. Typically, six categories are used for the level of educations, including primary school, junior high school, high school, university, vocational college, and oversea schools.

Given the dialog history and individual user characteristics, the main purpose of this article is to generate personalized responses.

B. Architecture of Our Approach

As shown in Fig. 1, the proposed MRPDG model jointly learns two tasks. The auxiliary task is author profiling,

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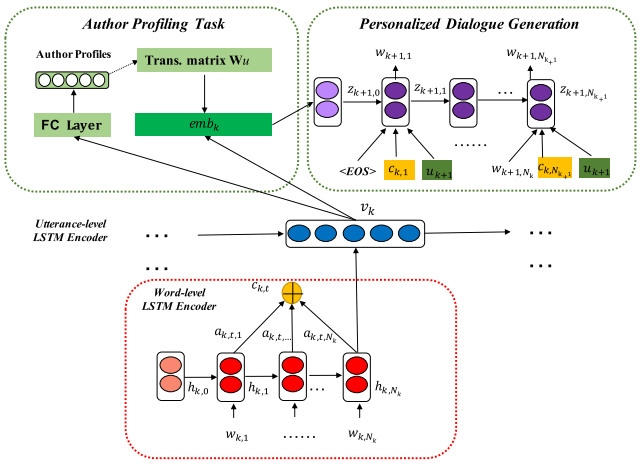

Fig. 1. Overview of the MRPDG model for a conversation composed of the utterances U_k and U_{k+1}. MRPDG is a multitask model consisting of two related tasks: author profiling (auxiliary task) and personalized dialog generation (primary task). Here, $w_{k,n}$ represents the nth input word in the utterance U_k, $\alpha_{k,t,n}$ represents the attention weight that captures the alignment between the nth position in utterance U_k and the tth position in predicted utterance U_{k+1}, $c_{k,t}$ is the context vector for generating the tth word in utterance U_{k+1}, and u_{k+1} denotes the user vector for U_{k+1}.

which predicts the user characteristics (i.e., age, gender, education, and location) and the primary task is personalized dialog generation. We use an LSTM encoder to learn utterance representations, which is shared by the personalized dialog generation and author profiling tasks. For personalized dialog generation, we first learn the user-specific sentence representations and then employ an LSTM decoder with the reinforcement learning techniques to generate personalized responses. Next, we will introduce the primary parts of our MRPDG model in detail.

IV. Author Profiling

Author profiling aims at ascertaining the user characteristics, such as age, country, gender, and job, by modeling their written texts. In this article, we focus on four kinds of user characteristics: *age*, *gender*, *education*, and *location*, which play a crucial role in personalized dialog systems. These four typical profiling tasks can be treated as four multiclass classification tasks, where each classifier takes as input the sentence and is optimized to predict which category the text sequence belongs to.

A. LSTM Encoder

1) Word-Level LSTM: We first convert the kth utterance $U_k = \{w_{k,1}, \ldots, w_{k,N_k}\}$ into a sequence of hidden states $H_k = \{h_{k,1}, \ldots, h_{k,N_k}\}$ by an LSTM encoder, where N_k is the length of the kth utterance. Specifically, we map each word $w_{k,t}$ at position t in utterance k to a distributed representation $e_{k,t} \in \mathbb{R}^d$ with an embedding layer, where d represents the dimensionality of the word embedding. Given the word embedding $e_{k,t}$ at position t in utterance k as input, the hidden state $h_{k,t}$ at time step t in utterance k is computed as

$$i_{k,t} = \sigma(W_i e_{k,t} + U_{k,t} h_{k,t-1} + b_i) \tag{1}$$

$$f_{k,t} = \sigma(W_f e_{k,t} + U_f h_{k,t-1} + b_f) \tag{2}$$

$$o_{k,t} = \sigma(W_o e_{k,t} + U_o h_{k,t-1} + b_o) \tag{3}$$

$$c_{k,t} = i_{k,t} \odot \tilde{c}_{k,t} + f_{k,t} \odot c_{k,t-1} \tag{4}$$

$$\tilde{c}_{k,t} = \tanh(W_c e_{k,t} + U_c h_{k,t-1} + b_c) \tag{5}$$

$$h_{k,t} = o_{k,t} \odot \tanh(c_{k,t}) \tag{6}$$

where c_t, i_t, f_t, and o_t are memory cell, input gate, forget gate, and output gate, respectively. U and W represent the learnable weight parameters. b indicates a set of bias terms. σ is a sigmoid function, and $\odot$ stands for the elementwise product. For simplicity, the calculation of the hidden state $h_{k,t}$ is recorded as $h_{k,t} = \text{LSTM}(h_{k,t-1}, w_{k,t})$. Hence, the initial utterance representation for U_k can be represented as $H_k = [h_{k,1}, h_{k,2}, \ldots, h_{k,N_k}]$, which is learned by the LSTM [32] encoder.

2) Utterance-Level LSTM: The utterance-level LSTM network tracks the conversation history by learning the dialog representation step by step iteratively. The utterance-level LSTM has a similar structure with the word-level LSTM. In particular, the hidden state of the utterance-level LSTM at kth utterance is v_k, which is computed as

$$v_k = \text{LSTM}(v_{k-1}, h_{k,N_k}) \tag{7}$$

where v_k summarizes the conversation history up to the kth utterance.

B. User Characteristic Prediction

The four typical profiling tasks are four multiclass text classification tasks. In particular, the utterance representation v_k for the kth utterance is passed to a task-specific fully connected layer and an output layer for age prediction given the kth utterance

$$\hat{y}_k^{\text{age}} = \text{softmax}(V_2 \cdot F_k) \tag{8}$$

$$F_k = \tanh(V_1 \cdot v_k) \tag{9}$$

where V_1 and V_2 represent learnable parameters. Similarly, we can get the predicted probability distribution over the categories for gender, education and location classification, denoted as $\hat{y}_k^{\text{gender}}$, $\hat{y}_k^{\text{edu.}}$, and $\hat{y}_k^{\text{loc.}}$, respectively.

The parameters of our model are optimized in a supervised manner. We assume that there are M labeled training instances, the objective function can be obtained by minimizing the cross-entropy between the ground-truth label distribution y and the predicted label distribution $\hat{y}$. Formally, the objective functions for the age, gender, education, and location classification are calculated as

$$L^{\text{age}} = -\frac{1}{M} \sum_{m=1}^{M} \sum_{i=1}^{K^{\text{age}}} \mathbf{I}\{y_m^{\text{age}} = i\} \log(\hat{y}_m^{\text{age}}) \tag{10}$$

$$L^{\text{gender}} = -\frac{1}{M} \sum_{m=1}^{M} \sum_{i=1}^{K^{\text{gender}}} \mathbf{I}\{y_m^{\text{gender}} = i\} \log(\hat{y}_m^{\text{gender}}) \tag{11}$$

$$L^{\text{edu.}} = -\frac{1}{M} \sum_{m=1}^{M} \sum_{i=1}^{K^{\text{edu.}}} \mathbf{I}\{y_m^{\text{edu.}} = i\} \log(\hat{y}_m^{\text{edu.}}) \tag{12}$$

$$L^{\text{loc.}} = -\frac{1}{M} \sum_{m=1}^{M} \sum_{i=1}^{K^{\text{loc.}}} \mathbf{I}\{y_m^{\text{loc.}} = i\} \log(\hat{y}_m^{\text{loc.}}) \tag{13}$$

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where $\mathbf{I}\{\cdot\}$ represents an indication function such that $\mathbf{I}\{\text{true}\} = 1$ and $\mathbf{I}\{\text{false}\} = 0$. K^{age}, K^{gender}, $K^{\text{edu.}}$, and $K^{\text{loc.}}$ denote the number of categories for age, gender, education, and location, respectively.

To learn a better semantic sentence representation, these four author profiling tasks are learned jointly. We define the joint objective function for the four author profiling tasks as

$$L_1 = L^{\text{gender}} + L^{\text{age}} + L^{\text{edu.}} + L^{\text{loc.}} \tag{14}$$

V. Personalized Dialog Generation

The personalized dialog generation subtask shares the sentence representation module (LSTM encoder) with the author profiling subtask. The next utterance U_{k+1} is generated by an LSTM decoder, which predicts the probability of each token in U_{k+1}.

A. User-Specific Sentence Representation

To generate the personalized response, the initial utterance representation is transformed to a user-specific utterance representation, which is expected to capture user characteristics. Inspired by [33], we develop a transformation operation to make the transformed utterance representation capture the user characteristics. In particular, MRPDG transforms the initial utterance representation of the kth utterance (i.e., v_k) to an age-specific utterance representation emb_k^{age} by

$$\text{emb}_k^{\text{age}} = \tanh\left(\mathbf{W}_{\mu}^{\text{age}} \times v_k\right) \tag{15}$$

where $\mathbf{W}_{\mu}^{\text{age}}$ is the transformation matrix. It is noteworthy that the initial utterance representation and the age-specific utterance representation have the same dimension.

To make the age-specific utterance representation aware of the user age information, the transformation matrix $\mathbf{W}_{\mu}^{\text{age}}$ is developed according to the predicted age information $\hat{y}_k^{\text{age}}$. We have K^{age} submatrices $(\mathbf{W}_{\mu,1}^{\text{age}}, \ldots, \mathbf{W}_{\mu,K^{\text{age}}}^{\text{age}})$, and each submatrix corresponds to one age label. Given the predicted age label predicted by (8), we calculate the transformation matrix $\mathbf{W}_{\mu}^{\text{age}}$ as the weighted sum of the submatrices: $\mathbf{W}_{\mu}^{\text{age}} = \sum_{i=1}^{|C|} \hat{y}_k^{\text{age}} \mathbf{W}_{\mu,i}^{\text{age}}$. Consequently, $\mathbf{W}_{\mu}^{\text{age}}$ is biased to the predicted age label.

Similarly, we can obtain the gender-specific utterance embedding $\text{emb}_k^{\text{gender}}$, education-specific utterance embedding $\text{emb}_k^{\text{edu.}}$, and location-specific utterance embedding $\text{emb}_k^{\text{loc.}}$. Finally, we employ a linear combination of the age-specific, gender-specific, education-specific, and location-specific utterance embeddings as the final user-specific utterance embedding for the kth utterance U_k

$$\text{emb}_k = \gamma_1 \text{emb}_k^{\text{age}} + \gamma_2 \text{emb}_k^{\text{gender}} + \gamma_3 \text{emb}_k^{\text{edu.}} + \gamma_4 \text{emb}_k^{\text{loc.}} \tag{16}$$

where we set $\gamma_1 = \gamma_2 = \gamma_3 = \gamma_4 = 0.25$, which means that these four kinds of user characteristics contribute equally to the final user-specific utterance representation.

B. LSTM Decoder

The LSTM decoder is, in essence, a neural language model except that it depends on the output of the LSTM encoder. We use the user-specific utterance representation emb_k of utterance U_k as the initial state of the LSTM decoder when generating the utterance U_{k+1}. At each decoding step t, the decoder takes $x_{k+1,t}$ (while training, $x_{k+1,t}$ is the embedding of the $t-1$th word in the ground truth utterance U_{k+1}; at test time, it is the embedding of the $t-1$th word emitted by the decoder) as input and compute the hidden state $z_{k+1,t}$ as

$$z_{k+1,t} = \text{LSTM}(z_{k+1,t-1}, x_{k+1,t}, c_{k+1,t}, u_{k+1}) \tag{17}$$

where $c_{k+1,t}$ represents a context vector at time step t, and u_{k+1} indicates the user vector. The context vector $c_{k+1,t}$ can be computed as the weighted sum of the hidden states H_k. Here, we compute the user representation u for each user by using the user characteristics (e.g., gender, location, age, education, and identity), and the learned user representation is fed into the decoder as an extra input. Finally, the hidden state $z_{k+1,t}$ is employed to generate personalized dialogs. We will present how we learn the context vector representation and user representation in the following.

1) Context Vector Representation: Following [12], the attention weights are used to determine the importance of the hidden representation H_k when decoding the response. The context vector $c_{k+1,t}$ is calculated as the weighted sum of the hidden representation of utterance U_k (i.e., H_k)

$$c_{k+1,t} = \sum_{n=1}^{N_k} \alpha_{k,t,n} h_{k,n}. \tag{18}$$

The attention weight $\alpha_{k,t,n}$ for each hidden state $h_{k,n}$ is calculated as

$$\alpha_{k,t,n} = \frac{\exp(e_{k,t,n})}{\sum_{n'=1}^{N_k} \exp(e_{k,t,n'})} \tag{19}$$

$$e_{k,t,n} = \sigma(z_{k+1,t-1}, h_{k,n}) \tag{20}$$

where σ is a fully connected layer that projects a vector to a real-valued attention weight. The attention weight $\alpha_{k,t,n}$ captures the alignment between the nth position of utterance U_k and tth position of the predicted utterance U_{k+1}. The attention weights allow the LSTM decoder to dynamically choose different parts of the utterance U_k.

2) User Representation: In dialog systems, user characteristics (e.g., gender, education, age, and job) play a crucial role in personalized dialog generation. In this article, each user is represented as a distributed embedding that encodes the user characteristics. We primarily consider the user identity and four kinds of user characteristics: *Age*, *Gender*, *Education*, and *Location*. We first transfer the user attributes into one-hot vector representations, which are denoted as u_i, u_a, u_g, u_e, and u_l, respectively. The user vector can be obtained by concatenating the attribute vectors: $u_{k+1}^e = [u_i, u_a, u_g, u_e, u_l]$. Since one-hot vector has the data sparsity problem, we then convert each one-hot vector to a low-dimensional user representation u by performing matrix-vector multiplication

$$u_{k+1} = W^e u_{k+1}^e \tag{21}$$

where W^e represents a learnable matrix parameter. The user representation u_{k+1} is then integrated into the LSTM decoder as the extra input.

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C. Dialog Generation With Policy Gradient

In this section, the policy gradient is employed to maximize the expected reward of the generated response, which is called MRPDG-PG. The policy gradient method attempts to follow the gradient of the expected reward by searching the policy space, which expresses stochastic optimal policies and is robust to small variations in the approximation of the value function. Therefore, under certain appropriate conditions, it is guaranteed that the policy gradient method will converge. Next, we will present the state, action, policy, and reward of the policy gradient method in detail.

1) State: For the $k+1$th utterance, the state s_{k+1} is represented by the conversation history (prior utterances) $U_{<k+1}$. To make use of the generalization ability of deep learning in feature learning, we employ the LSTM encoder to encode the conversation history $U_{<k+1}$ into a distributed state representation υ_k.

2) Policy: The stochastic policy gradient algorithm [18] is used to approximate the stochastic policy π_θ by applying a function approximator. In this article, the policy is represented by the seq2seq model (i.e., LSTM encoder–decoder network), denoted as $P(U_{k+1}|U_{<k+1};\theta)$. The input of the policy is the state representation, and its output is the probability of the selected action. Let θ denote the collection of the model parameters.

3) Action: An action a_{k+1} is the $k+1$th dialog utterance (U_{k+1}) to be generated by the policy $P(U_{k+1}|U_{<k+1};\theta)$. In dialog generation, the action space can be infinite as arbitrary-length sequences might be produced.

4) Reward: The policy gradient method optimizes the parametrized policy in regard to the expected reward by applying gradient descent. We assume that the proposed MRPDG-PG receives a reward r_{k+1} at the $k+1$th utterance. We introduce three primary components that constitute the expected reward in MRPDG-PG.

5) Topic Coherence: Participants in a dialog usually have an implicit goal to interact in a coherent way and would like to maintain an inherent topic in discussion. Inspired by the approach used in [34], we learn a topic vector for the conversation history by using latent Dirichlet allocation [35] method. Instead of feeding the learned topic representation into the seq2seq model directly as an extra input [34], we calculate the similarity between the topic representations of the generated response and the conversation history. The learned topic similarity is then used as the topic coherence reward for MRPDG. We use $\text{topic}_{<k+1}$ and topic_{k+1} to represent the topic vectors of the conversation history ($U_{<k+1}$) and the generated response (U_{k+1}). Formally, the topic coherence reward is computed as the cosine similarity between the two topic representations

$$\begin{aligned} r_{k+1}^{(\text{topic})} &= \log \text{cosine}(\text{topic}_{<k+1}, \text{topic}_{k+1}) \\ &= \log \frac{\text{topic}_{<k+1} \cdot \text{topic}_{k+1}}{\|\text{topic}_{<k+1}\| \|\text{topic}_{k+1}\|}. \end{aligned} \tag{22}$$

6) Mutual Information: Previous work [13] revealed the good performance of the mutual information in generating responses. In MRPDG, we use mutual information as a reward to estimate the semantic coherence between the previous utterance U_k and the generated response U_{k+1}. Formally, the weighted mutual information [13] between U_k and U_{k+1} is

$$r_{k+1}^{(\text{MI})} = (1-\omega)\log P(U_{k+1}|U_k) + \omega \log P(U_k|U_{k+1}) \tag{23}$$

where $p(U_{k+1}|U_k)$ indicates the probability of generating the response U_{k+1} conditioned on the prior utterance U_k. $p(U_k|U_{k+1})$ denotes the probability of generating the prior utterance U_k given the utterance U_{k+1}. ω denotes the hyperparameter that determines the weights of the two probabilities, and we set $\omega = 0.5$. $P(U_k|U_{k+1})$ and $P(U_{k+1}|U_k)$ are implemented using the same model, with U_k and U_{k+1} interchanged.

7) Language Model: To generate dialogs that are grammatical and fluent, we calculate a language model reward $r^{(\text{LM})}$, estimating how fluent and grammatical the generated response is. We compute the language model reward of the $k+1$th utterance generated by MRPDG as

$$r_{k+1}^{(\text{LM})} = \text{LM}(U_{k+1}) = \log P(U_{k+1}). \tag{24}$$

Following [36], the language model is implemented with the LSTM. We first pretrain the language model on Wikipedia corpora, and then use the pretrained language model to calculate the log-likelihood score that is used as the language model reward for the generated utterance.

Finally, we compute the total reward r_{k+1} for utterance U_{k+1} by linearly combining the topic coherence reward $r^{(\text{topic})}$, the mutual information reward $r^{(\text{MI})}$, and the language model reward $r^{(\text{LM})}$

$$r_{k+1} = \lambda_1 r_{k+1}^{(\text{topic})} + \lambda_2 r_{k+1}^{(\text{MI})} + \lambda_3 r_{k+1}^{(\text{LM})} \tag{25}$$

where we set $\lambda_1 = \lambda_2 = \lambda_3 = \frac{1}{3}$, which means that these three kinds of rewards contribute equally to the final rewards. Each reward is computed until reaching the last term of the generated utterance.

8) Estimating Model Parameters: We learn the policy $P(U_{k+1}|U_{<k+1})$ that results in the optimal expected reward $\mathbb{E}[r_k]$

$$L_2(\theta) = \mathbb{E}[r_{k+1}]. \tag{26}$$

In accordance with the policy gradient theorem [18], the gradient of the expected reward in response to the parameters θ of MRPDG-PG can be calculated as

$$\nabla_\theta L_2(\theta) = E[\nabla \log P(U_{k+1}|U_{<k+1}) \cdot (r_{k+1} - b_{\text{RL}})]. \tag{27}$$

We refer the interested reader to [18] and [38] for the implementation of policy gradient algorithm. b_{RL} represents a baseline strategy which is used to decline the learning variance, similar to [7] and [38]. The parameters of θ can be updated by employing stochastic gradient descent algorithm.

D. Dialog Generation With Deep Q-Network

In this section, the Q-learning method is employed to generate personalized dialogs, which is called MRPDG-QL. Q-learning is a value-based reinforcement learning method, aiming at learning an optimal action-value function (or Q value function) $Q(s,a)$. Given the state s and action a, the optimal

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Q value function $Q(s, a)$ is defined as the maximum expected reward. In practice, it is difficult to calculate the Q value for dialog generation, since the Q value depends on all potential state-action pairs that are infinite in generating arbitrary-length textual sequences. To overcome this issue, we can approximate the Q value function $Q(s, a)$ by employing parameterized function $Q(s, a; \theta)$.

Different from the standard Q-learning method, we employ the seq2seq model to approximate the Q value function $Q(s, a; \theta)$ (called deep Q-network). Concretely, we feed each hidden state of the LSTM decoder into a fully connected layer to approximate the Q value function.

Suppose that the optimal action-value function adheres to the Bellman equation [38]: if the optimal action-value function $Q(s', a'; \theta)$ of the generated utterance s' is known for all hypothetical actions, then the optimal policy is to choose an action a' by maximizing the expected value of $r+\lambda Q(s', a'; \theta)$ at each iteration i

$$Q(s, a; \theta_i) = \mathbb{E}[r_i + \lambda \max_{a'} Q(s', a'; \theta_{i-1}|s, a)] \tag{28}$$

where r_i denotes the current reward at iteration i defined in Section V-C via (25), λ denotes a discounted value for the future reward, θ_i (or θ_{i-1}) represents the parameters of the deep Q-network in ith (or $i-1$th) iteration.

We minimize the loss function $L_2(\theta_i)$, which changes in each iteration to optimize the Q-network

$$L_2(\theta_i) = \mathbb{E}_{s,a}[(y_i - Q(s, a; \theta_i))^2] \tag{29}$$

where $y_i = \mathbb{E}_{s,a}[r_i + \lambda \max_{a'} Q_i(s', a'; \theta_{i-1})|s, a]$ represents the target Q value with the parameters θ_{i-1} from the previous iteration. We update the parameters by differentiating the objective function with regard to the model parameters θ_i

$$\nabla_{\theta_i} L_2(\theta_i) = \mathbb{E}_{s,a}[2(y_i - Q(s, a; \theta_i))\nabla_{\theta_i} Q(s, a; \theta_i)]. \tag{30}$$

After obtaining the parameters of the model, we select the action using the ϵ-greedy approach. Specifically, we choose the action of the highest Q value $Q(s, a; \theta)$ with the probability of 1-ϵ and choose an arbitrary action with the probability of ϵ. The ϵ-greedy approach considers the trade-off between exploitation and exploration.

E. Dialog Generation With Actor-Critic Method

In this section, the AC method is employed for personalized dialog generation, which is called MRPDG-AC. The AC method is a widely used reinforcement learning algorithm, combining the advantages of the value-based and policy-based techniques. Given the policy of the *actor*, the *critic* uses an LSTM network to approximate the value function. The *actor* utilizes this approximation of the value function to update its parameters in the direction of improving the expected reward.

Similar to the deep Q-network, we train the *critic* by minimizing the squared temporal difference error (loss function) at each iteration

$$L_2(\theta) = \mathbb{E}[((r_{k+1} + \lambda Q(s_{k+1}, \pi'(s_{k+1}; \phi'); \sigma') - Q(s_k, a_k; \sigma))^2] \tag{31}$$

where Q represents the Q-network, λ denotes a discount factor, $\pi'(s_{k+1})$ denotes the target *actor* for the $k+1$th utterance, and σ (or σ') represents the parameter set of the *critic* (or target *critic*), ϕ' denotes the parameters of the target *actor*. The parameters of the *critic* can be updated using gradient descent algorithm.

We calculate the gradient of the loss functions in response to the parameters by backpropagating the gradient of the value function in regard to the actions taken by the *actor*, respectively. Following the idea of [39], the *actor* is trained by exploiting the gradient of the deterministic policy:

$$\nabla_\phi \pi(s_t, a_t) = \mathbb{E}[\nabla_{a_t} Q(s_t, a_t; \sigma)\nabla \pi_\phi(s_t; \phi)]. \tag{32}$$

For detailed implementation, we refer the reader to [19] and [41].

Following the ideas proposed in [19], we use attention-based LSTM encoder–decoder model to implement the *critic* and the *actor*. During the training stage, the *actor* takes as input the utterance U_k and generates the target utterance U_{k+1} that is judged by the *critic* network. The *critic* receives the ground-truth utterance U_{k+1} as the input of the LSTM encoder and receives the token $w_{k+1,t-1}$ predicted by the actor network as input at time step t of the LSTM decoder. The value function Q learned by the *critic* network is employed to calculate the gradient of the expected rewards in regard to the parameters of the *actor* network.

F. Multitask Learning

Overall, MRPDG has two subtasks, and each has a learning objective. To make the sentence representation aware of the user characteristics, we learn these two related tasks jointly. The joint objective function for MRPDG can be minimized by

$$L = \beta_1 L_1 + \beta_2 L_2 \tag{33}$$

where β_1 and β_2 are hyperparameters that controls the importance of L_1 and L_2. In the experiments, we set $\beta_1 = \beta_2 = 0.5$. We use the minibatch ADADELTA optimization algorithm [41] to update the parameters of the model.

VI. Experimental Setup

A. Data Sets Description

To evaluate the effectiveness of MRPDG, extensive experiments are conducted on two dialog data sets. We describe the properties of these two data sets as follows:

1) Switchboard Dialogue Act Corpus in English: The switchboard dialogue act (SwDA) data are extension of the switchboard-1 telephone speech corpus.[1] In SwDA, each utterance has the user profile information (e.g., "from-caller-no," "from-caller-gender," "from-caller-education," "from-caller-birth-year," "from-caller-dialect-area," "to-caller-no," "to-caller-gender," "to-caller-education," "to-caller-birth-year," and "to-caller-dialect-area"). Similar to [11], SwDA is divided into the training data (from folder sw00 to folder sw09), the validation data (from folder sw11 to folder sw13), and the test data (folder sw10). Specifically, SwDA consists

[1] http://compprag.christopherpotts.net/swda.html

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of 190 000 utterances (98 700 turns) for training, 11 400 utterances (5 700 turns) for validation, and 22 200 utterances (11 900 turns) for testing, respectively.

2) Sina Data in Chinese (Denoted as Sina): We collect a large-scale Chinese data set from a popular Chinese microblogging website Sina Weibo[2], to evaluate the proposed MRPDG in generating personalized dialogs. Similar to the strategy utilized in [3], we first collect a large number of postresponse pairs, and then untangle the data from the crawled raw data by using a similar strategy as suggested in [42], such as filtering out the inconsequential responses (e.g., "wow") and advertisements. Concretely, we collect 251 921 postresponse pairs from Sina Weibo as our experimental data (Sina data set). From the Sina data set, 1000 instances are then randomly selected as our validation and test data, respectively, and the remaining instances are utilized for training the model. We also crawled user characteristics information, e.g., user id, user description, city ID, province ID, gender, age, and education, for each involved user. The Sina data contains the conversations with one round, in which each conversation is composed of a post (the input of the dialog system) and a response (the output of the dialog system).

We perform data preprocessing before conducting experiments. For the SwDA data set, each text is tokenized using a widely used text processing toolkit NLTK.[3] We build two independent vocabularies for posts and responses by keeping the top 10 000 words with the highest frequency. The rest tokens that are not contained in the vocabularies are displaced by an "*UNK*" token.

B. Implementation Details

We utilize 300-D word2vec [43] trained on English and Chinese Wikipedia corpora to initialize the word embeddings in English and Chinese data sets, respectively. We fine-tune the word embeddings when training the model. For both data sets, the recurrent weight parameters are initialized as orthogonal matrices. The other weight parameters are initialized with the normal distribution $\mathcal{N}(0, 0.01)$, and the bias terms are initialized to zero. We use the ADADELTA [41] optimization method to optimize the proposed MRPDG. We set the learning rate to 1×10^{-4}, and the discount factor λ to 0.95. Batch size is set as 64. L_2 regularization (weight decay $= 0.001$) and dropout strategy (dropout rate $= 0.2$) are employed to alleviate the overfitting problem.

Following the method proposed in [7], the proposed MRPDG model is pretrained on the OpenSubtitles data set [44] with both English and Chinese files in a supervised manner so as to learn good general conversation patterns.

C. Compared Methods

Several state-of-the-art baseline methods are adopted for comparison, which we describe in the following.

1) *Seq2Seq:* This model applies a four-layer LSTM to project the given utterance into an embedding with the fixed size. Another four-layer LSTM is then used to generate the target response [45]. The weight parameters of the LSTM networks are initialized with the uniform distribution $[-0.1, 0.1]$. We optimize the model by employing the ADATELTA optimization algorithm.
2) *Neural Responding Machine (NRM):* NRM employs the seq2seq model with an attention mechanism to produce responses given the posts [3]. The weight parameters of NRM are initialized with the uniform distribution $[-0.1, 0.1]$. ADATELTA optimization algorithm is used to train the model by conducting mini-batch training (with batch size 64).
3) *Hierarchical Recurrent Encoder-Decode Model:* It extends the standard seq2seq model by exploring two-level hierarchy: word level and utterance level [2]. We set the size of hidden states for the encoder LSTM, context LSTM, and decoder LSTM to 300. The dropout rate of 0.2 is adopted.
4) *Deep Reinforcement Learning (DRL):* DRL [7] uses the policy gradient algorithm to reward the high-quality dialogs. We initialize the word embeddings with 300-D word2vec vectors. The sizes of the hidden states of the LSTM encoder and decoder are set to 300. ADATELTA optimization method is used to train the model by conducting mini-batch training (with batch size 64).
5) *Speaker–Addressee Model:* SAM encodes the personas into distributed representations and proposes an SAM to model the properties and interactions of the users [4]. We implement this model with the released code.[4] We initialize the parameters with the uniform distribution $[-0.1, 0.1]$. The dropout rate is set to 0.2.
6) *Personalized Responding Model:* It is an extension of the standard seq2seq model, which learns the specific speaking style of users by using a simple domain adaptation method [1]: pretraining and adaptation. The sizes of the hidden states of the LSTMs are set to 300. ADATELTA optimizer is used to train the model. The batch size is set to 64.
7) *Personalized Response Generation via Domain Adaptation (PRG-DA):* This model employs the domain adaptation strategy to learn personalized responses, which first pre-trains the model with a large amount of dialog data, and then fine-tunes the model with a small-scale personalized data by employing the dual-learning algorithm [24]. We initialize the word embeddings with 300-D word2vec vectors that are trained on the dumps of the English and Chinese Wikipedia. We initialize the parameters of the LSTM units with orthogonal matrices. The sizes of the hidden states of the LSTMs are set to 300. ADATELTA optimizer is used to train the model. The batch size is set to 64.

It is noteworthy that, for all the compared methods, we choose the best hyperparameters of the models with the validation data.

[2]http://weibo.com
[3]http://www.nltk.org
[4]https://github.com/jiweil/Neural-Dialog-Generation

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TABLE I

EXPERIMENTAL RESULTS IN TERMS OF BLEU. VALUES WITH * INDICATE THAT THE IMPROVEMENT FROM MRPDG IS STATISTICALLY SIGNIFICANT OVER THE COMPARED MODELS (T-TEST AND P-VALUE < 0.05)

Method	SwDA	Sina
Seq2Seq	1.60%	1.12%
NRM	1.73%	1.19%
HRED	1.71%	1.21%
DRL	1.75%	1.25%
SAM	1.78%	1.24%
PRM	1.81%	1.29%
PRG-DA	1.87%	1.30%
MRPDG-PG	1.99%*	1.43%*
MRPDG-QL	1.91%	1.37%
MRPDG-AC	2.04%*	1.49%*

TABLE II

EXPERIMENTAL RESULTS IN TERMS OF PERPLEXITY

Method	SwDA	Sina
Seq2Seq	62.5	42.2
NRM	60.5	41.0
HRED	60.9	40.5
DRL	59.3	40.6
SAM	61.1	39.9
PRM	60.3	40.2
PRG-DA	58.9	39.5
MRPDG-PG	57.9*	37.8*
MRPDG-QL	58.2	38.3*
MRPDG-AC	57.1*	36.8*

VII. EXPERIMENTAL RESULTS

A. Automatic Evaluation

Evaluating open-domain dialog systems is challenging. We adopt similar automatic metrics as in [4]. Concretely, the proposed MRPDG is compared with the baseline methods over two automatic evaluation metrics (i.e., BLEU and perplexity).

1) BLEU Evaluation: BLEU [5] calculates n-gram overlaps between the generated response U and the ground-truth response $\hat{U}$, where the overlaps are weighted by n-gram rarity and saliency. Formally, the n-gram precision for the generated response is computed as

$$P_n(U, \hat{U}) = \frac{\sum_k \min(\eta(k, U), \eta(k, \hat{U}))}{\sum_k \eta(k, U)} \tag{34}$$

where k traverses all candidate n-grams, U and $\hat{U}$ represent the ground-truth and generated utterances, and $\eta(k, U)$ indicates the number of n-grams in U. After obtaining the precision, BLEU-N (N represents the maximum length of possible n-grams) is then computed as

$$\text{BLEU-}N = b(U, \hat{U}) \exp\left(\sum_{n=1}^{N} \beta_n \log P_n(U, \hat{U})\right) \tag{35}$$

where $\beta_n = 1/N$ is a weight score and the most widely adopted version of BLEU chooses $N = 4$. $b(U, \hat{U})$ is a brevity penalty that penalizes short sentences

$$b(U, \hat{U}) = \begin{cases} 1, & \text{if } |\hat{U}| > |U| \\ e^{(1-|U|/|\hat{U}|)}, & \text{otherwise.} \end{cases} \tag{36}$$

The higher the BLEU score, the better the model.

The BLEU scores on SwDA and Sina data sets are reported in Table I. We observe that the BLEU scores of the proposed methods (i.e., MRPDG-PG, MRPDG-QL, and MRPDG-AC) stably outperform the compared methods by a large margin on SwDA and Sina data sets, and MRPDG-AC achieves the best results on both experimental data sets. Values with * indicate that the improvement from MRPDG is statistically significant over the compared models (t-test, p-value < 0.05). Specifically, for SwDA data, MRPDG-AC achieves the best BLEU value of 2.04%, which is 0.17% higher than the best score of the compared approaches (i.e., PRG-DA) (with 9.09% improvement). Similar trends are observed on the Sina data set. MRPDG-AC improves about 14.6% over the compared models. The main advantage of MRPDG-AC over MRPDG-PG and MRPDG-QL comes from its capability of exploiting the strong points of both the value-based and policy-based methods, which represents the policy independently of the value function. MRPDG-QL performs worse than MRPDG-PG, since MRPDG-QL uses a deep neural network (LSTM) to approximate the Q-values, and small updates to the approximated Q value may significantly change the policy and the data distribution. Rather than learning an approximation of the value function as in MRPDG-QL, MRPDG-PG tries to search the policy space directly, which is robust to slight changes in value function approximation. Thus, MRPDG-PG achieves better performance than MRPDG-QL.

2) Perplexity Evaluation: We also evaluate the effectiveness of MRPDG with perplexity that is widely used to measure how well the model predicts a sequence. For an utterance $U = \{w_1, w_2, \ldots, w_l\}$, its test perplexity by a model trained with the document set D is defined as

$$H(U, D) = -\frac{1}{l} \log_2 P_D(w_1, w_2, \ldots, w_l) \tag{37}$$

$$\text{Perplexity}(U, D) = 2^{H(U,D)} \tag{38}$$

where P_D is a probability computed by the generative model trained with D. $H(U, D)$ is the entropy of utterance U given the document set D. The lower the perplexity, the better the model.

Similar to the experiments of [2], we compute the average log-probability for the last turn of utterance in the dialog. The perplexity results on SwDA and Sina data sets are reported in Table II. As shown in Table II, the perplexity of MRPDG-AC decreases from 58.9 and 39.5 (best results from PRG-DA) to 57.1 and 36.8 over SwDA and Sina data sets, respectively. MRPDG performs statistically significantly better than the compared models.

3) Learning Curve of Our Model: To better understand the training process of MRPDG, we illustrate the learning curves of MRPDG in response to the BLEU and perplexity values. Figs. 2 and 3 show the learning curves of the automatic evaluation metrics during the reinforcement learning process for the proposed models on SwDA and Sina data sets, respectively.

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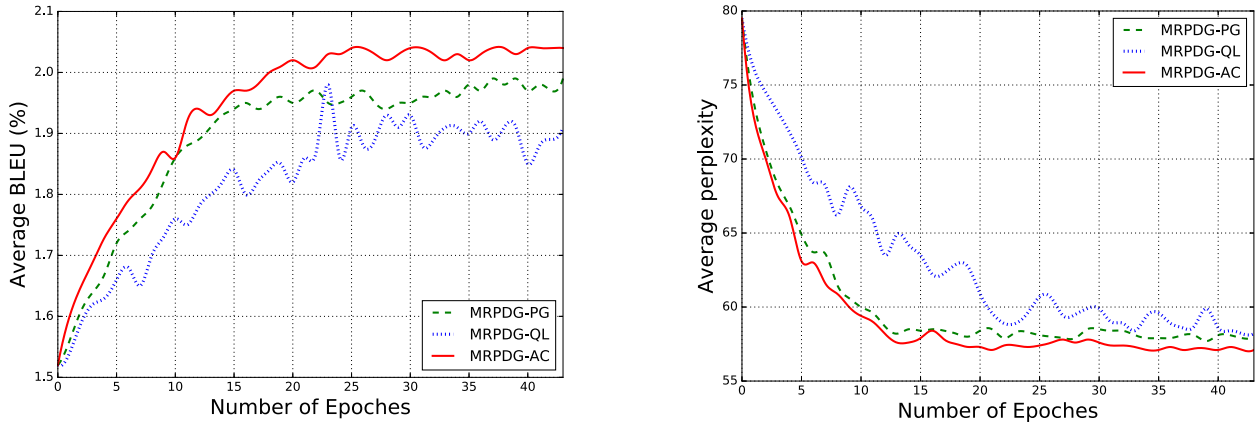
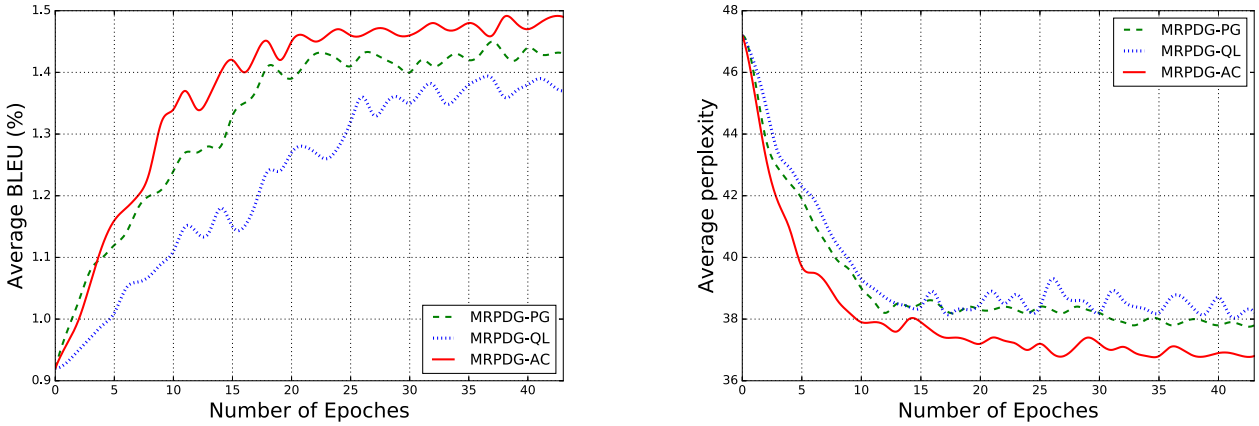


(a) (b)

Fig. 2. Learning curve of the three proposed models w.r.t. (a) BLEU and (b) perplexity on SwDA data set.



(a) (b)

Fig. 3. Learning curve of the three proposed models w.r.t. (a) BLEU and (b) perplexity on Sina data set.

As shown in Fig. 2, on the SwDA data set, MRPDG-PG and MRPDG-AC quickly converge to the optimal values after about 20 epochs for BLEU and 15 epochs for perplexity, while MRPDG-QL converges more slowly. This is because the value-based reinforcement learning (i.e., MRPDG-QL) distinguishes the policy actions when changing the value function slightly. Specifically, small changes in the estimated value function may affect the predicted action significantly, which is also related to the instability of MRPDG-QL. The policy-based method (i.e., MRPDG-PG) works directly in policy space and has much smoother learning curves and a performance improvement guarantee with each update. However, the MRPDG-PG method tends to converge to local optima. The AC method (i.e., MRPDG-AC) combines the advantages of both value-based and policy-based methods. This allows MRPDG-AC to be more efficient via TD updates at every step and achieves the best results in personalized dialog generation. From Fig. 3, similar trends are observed on the Sina data set. MRPDG-PG and MRPDG-AC converge to the optimal BLEU after about 20 epochs, while MRPDG-QL converges after about 25 epochs.

4) Ablation Study: We also perform ablation study of the proposed models by removing author profiling task (w/o AP), reinforcement learning algorithm (w/o RL), topic reward (w/o topic), language model reward (w/o LM), and mutual information reward (w/o MI), respectively. It is noteworthy that without using reinforcement learning to train the MRPDG model, the topic reward cannot be integrated into the seq2seq framework because of the nondifferentiable issue of the MLE training. Thus, the learning objective for the MLE training (w/o RL) contains three parts: the probability of generating the target response, the language model probability, and the mutual information probability. We repeat the experiments three times with the different random seed values in the experiments. Due to the limited space, we only report the average results of MRPDG-AC (our best model). The results are illustrated in Tables III and IV for SwDA and Sina data sets, respectively. In general, all the rewards contribute, but the author profiling and reinforcement learning contribute the most. This is within our expectation since: 1) the author profiling task can help capture user characteristics, e.g., speaking style and background information, and guide the MRPDG model to

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TABLE III
ABLATION STUDY FOR MRPDG-AC ON SWDA DATA

Method	BLEU	Perplexity
MRPDG-AC	2.04%	57.1
w/o AP	1.77%	60.5
w/o RL	1.74%	60.8
w/o topic	1.89%	57.9
w/o MI	1.85%	59.5
w/o LM	1.93%	58.3

TABLE IV
ABLATION STUDY FOR MRPDG-AC ON SINA DATA

Method	BLEU	Perplexity
MRPDG-AC	1.49%	36.8
w/o AP	1.18%	39.2
w/o RL	1.21%	39.8
w/o topic	1.33%	37.9
w/o MI	1.28%	38.5
w/o LM	1.31%	37.7

locate the user-specific information in a dialog and 2) the reinforcement learning can overcome the nondifferentiable and "exposure bias" problems, enabling the model to produce fluent, coherent, and informative responses by maximizing the expected reward over the conversation.

B. Human Evaluation

The automatic evaluation metrics perplexity and BLEU have wide adoption in evaluating dialog systems. However, how well these automatic metrics are matched with human judgment is still controversial. Since the aim of the dialog systems is to generate the dialogs with long-term success, rather than predicting the responses with the highest probability, we also use human annotation to evaluate the dialog systems quantitatively. In particular, for SwDA data, we select 100 conversations produced by the models trained on SwDA data. For Sina data, we choose 100 posts from the test set and produce the response for each post. We invite three human annotators to score the generated responses based on their fitness (whether the generated response is coherent and natural given the dialog history) and fluency (whether the generated response is fluent with a proper grammatical structure). The annotators are invited to independently assign each response a score of 0 (bad), 1 (not bad), and 2 (good) for fitness and fluency, respectively. We report the average rating scores over the generated responses from all annotators as the final human evaluation results.

The human evaluation results are reported in Table V. MRPDG-AC consistently and substantially outperforms the compared approaches by a noticeable margin on both data sets. For instance, the *Fitness* scores of MRPDG-AC are 0.15 and 0.19 higher than the best scores of the compared methods (i.e., PRG-DA) on SwDA and Sina, respectively. The main advantage of MRPDG-AC comes from its ability to exploiting user characteristics and integrating reinforcement learning into the seq2seq framework.

TABLE V
HUMAN EVALUATION SCORES IN TERMS OF *Fitness* AND *Fluency* FOR BASELINE METHODS AND OUR MODELS

Method	SwDA dataset		Sina dataset	
	Fitness	Fluency	Fitness	Fluency
Seq2Seq	0.88	0.96	0.76	0.92
NRM	0.93	1.01	0.79	0.98
HRED	0.97	1.08	0.83	1.03
DRL	0.95	1.15	0.91	1.12
SAM	1.05	1.13	0.92	1.08
PRM	1.02	1.21	0.85	0.99
PRG-DA	1.14	1.24	0.98	1.15
MRPDG-PG	1.26*	1.31*	1.12*	1.18
MRPDG-QL	1.16	1.28*	1.05*	1.09
MRPDG-AC	1.29*	1.36*	1.17*	1.23*

C. Qualitative Evaluation

We conduct extensive experiments to estimate MRPDG qualitatively. Some output responses by MRPDG-AC and the compared models (DRL and PRG-DA) are demonstrated. Due to the space limitation, we select several conversations produced by DRL (the best dialog generation model without considering user characteristics), PRG-DA (the best personalized dialog generation model), and MRPDG-AC for comparison. For the SwDA data set, we demonstrate two conversations within five turns. Specifically, the first turn (i.e., Post A) is input by the authors, and two agents implemented with the same model (DRL, PRG-DA, or MRPDG-AC) take turns conversing, taking the previous utterance generated by the other agent as input. Table VI reports the conversations generated by DRL, PRG-DA, and MRPDG-AC, respectively. From the results, we observe that MRPDG-AC is prone to generating more forward-looking utterances (e.g., "How about you?" and "We shall have a party"). For the Sina data set, we choose five users from the test set and incorporate their characteristics information into DRL, PRG-DA, and MRPDG-AC models, respectively. The generated personalized responses in response to the given question are reported in Table VII. MRPDG-AC has a tendency to generate meaningful and specific responses for different users with respect to the given question. In particular, MRPDG-AC gives appropriate but quite distinct responses, with different perspectives and choices of words. For example, MRPDG-AC generates "加班太累了, 求安慰。(*working overtime is too tiring, seeking comfort.)*", where the response indicates that the user of response 1 was working overtime during the holidays. This verifies the effectiveness of our MRPDG-AC model in capturing user characteristics. By contrast, DRL tends to generate general responses and barely capture the details.

We also conduct experiments to measure the consistency of the conversations generated by DRL, PRG-DA, and MRPDG-AC, respectively. Each conversation is associated with a specified user ID. Similar to the strategy used in [4], we collected twelve pairs of questions, and it is expected

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TABLE VI
TWO EXAMPLE DIALOGS GENERATED BY DRL, PRG-DA, AND MRPDG-AC ON TWITTER DATA

Model	Dialogue 1	Dialogue 2
Post A	"Where do you live?"	"What do you like to cook?"
1st utterance by DRL	"I'm Dallas."	"I like to cook."
2nd utterance by DRL	"really?"	"I have. have you?"
3rd utterance by DRL	"yes, you will?"	"Yeah"
4th utterance by DRL	"all right"	"uh, I don't know"
Post A	"Where do you live?"	"What do you like to cook?"
1st utterance by PRG-DA	"I lived in New York."	"I like Italian food."
2nd utterance by PRG-DA	"I love Chicago."	"I like to cook Italian food."
3rd utterance by PRG-DA	"that's good."	"We have a party."
4th utterance by PRG-DA	"yes."	"It's good."
Post A	"Where do you live?"	"What do you like to cook?"
1st utterance by MRPDG-AC	"I lived in New York."	"Italian food. How about you?"
2nd utterance by MRPDG-AC	"I love New York. "	"I love India food."
3rd utterance by MRPDG-AC	"yeah, how about you?"	"That's good."
4th utterance by MRPDG-AC	"I am from Los Angeles."	"Yeah. We shall have a party."

TABLE VII
FIVE EXAMPLE CONVERSATIONS IN RESPONSE TO FIVE DIFFERENT USERS GENERATED BY DRL, PRG-DA, AND MRPDG-AC ON SINA DATA SET

Dialogue	DRL	PRG-DA	MRPDG-AC
Post	"假期第二天，你们都去哪儿玩了？" ("Where did you go on the second day of holiday?")	"假期第二天，你们都去哪儿玩了？" ("Where did you go on the second day of holiday?")	"假期第二天，你们都去哪儿玩了？" ("Where did you go on the second day of holiday?")
Resp. 1	"早点休息完 " ("Go to bed early")	" 每天加班很辛苦啊" ("Working overtime everyday is so hard.")	"加班太累了，求安慰 " ("Working overtime is too tiring, seeking comfort.")
Resp. 2	"朋友圈" ("Circle of friends")	" 要好好学习，有信心 " ("Study hard and have confidence.")	"苦逼的我只能在家学习" ("I can only learn at home.")
Resp. 3	" 假期愉快，节日快乐" ("Happy holidays.")	" 哈哈，我们去旅游吧 " ("Ha-ha, let's go to travel.")	"去了杭州一日游，你呢？" ("I went to Hangzhou for a day trip. How about you?")
Resp. 4	" 好的" ("Like eating.")	" 喜欢好吃的啊 " ("Good.")	" 放假要多吃点 " ("Eat more during the holidays.")
Resp. 5	" 我今天要上班 " ("I have to work today.")	" 感觉累并快乐. " ("I am tired and happy.")	" 没有假期，在上班呢" ("No holidays, have to work.")

TABLE VIII
EXAMPLES OF SPEAKER CONSISTENCY GENERATED BY DRL, PRG-DA, AND MRPDG-AC ON SINA DATA SET

	DRL	PRG-DA	MRPDG-AC
Question 1	"你住在哪个国家?" "What country do you live in?"	"你住在哪个国家?" "What country do you live in?"	"你住在哪个国家？" "What country do you live in?"
Response 1	"我也来自中国." "I'm also from China."	"我来自中国." "I'm from China."	"我住在中国." "I'm in China."
Question 2	"你来自哪里?" "Where are you from?"	"你来自哪里?" "Where are you from?"	"你来自哪里?" "Where are you from?"
Response 2	"我不. 在哪里呢？" "I don't. Where?"	"来自中国." "I'm from China."	"我来自中国. 你呢？" "I'm from China. And you?"
Question 3	"你多大了?" "How old are you?"	"你多大了?" "How old are you?"	"你多大了?" "How old are you?"
Response 3	"我19岁大" "I'm 19."	"我21岁" "I'm 21 years old."	"我21岁" "I'm 21."
Question 4	"你年龄是多少?" "What is your age?"	"你年龄是多少?" "What is your age?"	"你年龄是多少?" "What is your age?"
Response 4	"我不认为25岁" "I don't think 25."	"21岁" "21 years old."	"我21岁" "I'm 21."

that the responses for each pair of questions should be consistent if the dialog systems are personalized and coherent. For instance, the responses to "你住在哪个国家? (*what country do you live in?)*" and "你来自哪里？(*where are you from?)*" should be consistent given the same user characteristics. Table VIII shows the consistency of the produced conversations in response to two users chosen from the Sina test set. Both MRPDG-AC and PRG-DA can generate consistent responses given similar questions regarding the same user. The key reason for the good performance of MRPDG-AC

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and PRG-DA is to incorporate user information into the encoder–decoder framework. In addition, MRPDG-AC can generate more interactive responses than PRG-DA by employing reinforcement learning techniques to optimize long-term conversation rewards.

VIII. Conclusion

In this article, we proposed a personalized dialog system that produced better personalized responses than the compared models. Our model jointly trained the personalized dialog generation with the related author profiling task (auxiliary task), which helped to learn better user-specific dialog representations. In addition, we investigated the performance of three widely used reinforcement learning methods: Q-learning, policy gradient, and AC methods. Extensive experiments were conducted to measure the effectiveness of MRPDG. Experimental results showed that MRPDG achieved better BLEU, perplexity, and human evaluation results than the compared methods. In addition, the qualitative evaluation also verified that MRPDG was capable of producing more coherent personalized dialogs in response to different users than the compared methods.

It is noteworthy that automatic evaluation metrics, such as BLEU, are often not optimal and do not work very well on this task, especially when the reference responses are short. As shown in Table I, the average BLEU scores are less than 3% for all models discussed in this article. In the future, we will explore automatic evaluation metrics that may better match human judgments. In addition, we also plan to adopt the proposed MRPDG model to the task-oriented dialog systems. Last but not least, we will consider integrating the sentiment signals of users into dialog systems to generate responses with specific sentiments.

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